23244

ASSIGNMENT NO. 11

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Batch : G-10

Date of performance: 1-10-2020
Date of submission: 4-12-2020

Title: Strategy design pattern

Aim: Implement and apply strategy design pattern for simple shopping cart where three payment strategies are used such as credit card, PayPal, BitCoin. Create an interface for strategy pattern and give concrete implementation for payment.

Objective: To learn the concept of strategy design pattern.

Theory:

1) Strategy design pattern:

- In this, a class behaviour or its algorithm can be changed at runtime.
- We create objects which represent various strategies and a context object whose behaviour varies as per its strategy object.
- This strategy object changes the executing algorithm of the context object.

2) Design pattern representation:

- It represents the best practices used by experienced object - oriented software developers.

Creational design patterns:
- Abstract factory : creates instance of several family of classes.
- Builder : Seperates object construction from representation.
- Factory method : creates instance of several derived classes.
- Object pool : Recycling objects that are no longer in use.

Structural design patterns:
- Adapter : Match interfaces of different classes.
- Bridge : Seperates an object interface from its implementation.
- Decorator : Add responsibilities ~~of~~ to code dynamically.
- Private class data : Restricts accessor access.

3) Intent of design pattern:

- A design pattern systematically names and explains a general design that addresses a reccuring design problem.
- Intent is to present in a consistent and conherent manner the solution to a recurring design problem.

Class diagram:
Strategy design pattern:

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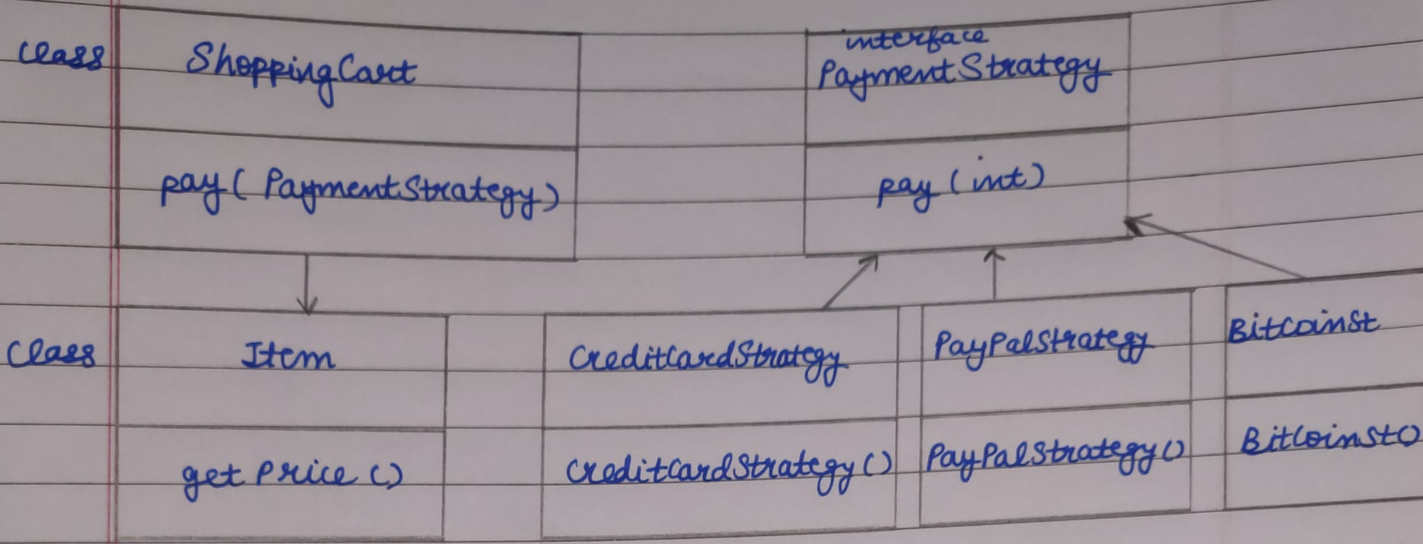

Advantages:

- It is easy to switch between different algorithms.
- clean code, because you avoid conditional - infested code.
- A class to each strategy, so we seperate the concers into classes.
- Can swap between algorithms during runtime.
- Replaces inheritance with composition.

Disadvantages:

- Mostey the algorithms are same for different strategies so, there is no need to complicate code unnecessarily.
- Clients must be aware of difference between stratgies to select a proper one.

Applications:

- Email implementation.
- File storage implementation